Investments: Assignment

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Instructions: This is a take-home group assignment. All relevant information can be found on Toledo under content *Assignment*:

1. Team groups: AssignmentTeams2026.xlsx
2. Problem setting: Assignment2026.pdf
3. Data to solve the assignment: DATA2026.xlsx

Write a detailed report to explain, step-by-step, the solutions you obtain: clearly present and interpret the results. This also includes explanatory notes on any underlying assumptions that you make. Do not round any intermediate results and report the solutions with 4 decimals. This assignment counts for 20% of the total examination score.

AI-tools can be used in a responsible way, in line with KU Leuven's guidelines for students on the authorised use of GenAI. Be transparent about your use by completing a transparency statement, explaining which tools have been used, and how the tools have been used.

Upload your report, including the transparency statement, in pdf format (no excel file/Pyton code should be added) on Toledo via content *Assignment* by the deadline of December 15, 2026, 1.00pm.

Good luck!

You have recently joined the active equity team of an asset management firm. Most of the firm's US equity portfolio is invested passively in a broad market portfolio represented by the MSCI US Index. However, the firm also employs a team of equity analysts who conduct fundamental research on individual companies with the aim of identifying stocks that are expected to outperform or underperform the market.

For the coming year, the firm's strategists expect the MSCI US Index to earn a return of 9.75% per annum. At the same time, funds can be invested in a risk-free 3-month U.S. Treasury bill yielding 4.2% per annum.

The equity research team has recently completed an in-depth analysis of 10 US stocks. Based on earnings forecasts, valuation models, industry developments, and company-specific information, the analysts have produced estimates of each stock's alpha, expressed as an expected abnormal return in % per annum:

Stock	**Alpha return (% p.a.)**
3M (MMM)	1.25
Alphabet A (GOOGL)	-0.50
American Express (AXP)	-1.00
Apple (AAPL)	0.90
Chevron (CVX)	0.00
Cisco Systems (CSCO)	-0.80
Honeywell Intl. (HON)	1.25
Travellers Cos. (TRV)	0.50
Walmart (WMT)	1.00

Your task is to determine how the firm can use these research insights to construct an active-passive portfolio that improves upon a purely passive investment strategy. That is, rather than investing all available wealth in the MSCI US Index, you will investigate whether taking active positions in the 10 researched stocks, in addition to the MSCI US Index, can improve the portfolio's risk-return profile. In constructing your portfolio, you cannot short sell.

To to do this, you have, in addition to the analysts' alpha forecasts, historical observations of the daily total return indices of the 10 individual stocks and the MSCI US Index. You also have historical data of the 3-month US Treasury bill rate, expressed in % per annum. Use these data to estimate the stocks' market beta exposures and to construct (expected) excess returns and the corresponding covariance matrix. For consistency with the alpha forecasts, express all estimates from the single index model on an annual

basis. Assume 250 trading days per year for stock returns. The Treasury bill rates may be converted to daily APRs assuming a 360-day year.

1. What are the expected excess returns of each of the investable assets?

2. What are the variances, systematic variances and idiosyncratic variances of each of the investable assets?

3. Determine the minimum variance frontier and the global minimum variance portfolio; use 21 portfolio combinations to construct the frontier. Do this in the expected excess return vs risk plane.

4. What are the Sharpe ratio, expected excess return, risk and composition of the optimal risky portfolio calculated via a maximization procedure.

5. What are the Sharpe ratio, expected excess return, risk and composition of the optimal risky portfolio calculated via the closed-form step-by-step procedure?

6. Determine the capital allocation line of the tangency portfolio, also in the excess return vs risk plane.

7. What is the optimal complete portfolio assuming a risk aversion parameter of 6. Determine its expected excess return, risk and composition.

8. For which risk aversion parameter will an investor want to leverage the tangency portfolio?

9. On an expected excess return-risk diagram plot the different investable assets, the minimum variance frontier, the global minimum variance portfolio, the tangency portfolio with capital allocation line, and the optimal complete portfolio assuming the risk aversion parameter of 6. Label all elements clearly.

10. Since the equity analysts are expensive, the asset management firm will want to charge a fee to its investors in the active-passive fund. How large can the annual management fee be (expressed as a percentage) to justify the additional cost of active management as compared to a no-fee passive MSCI US Index investment.